EXP.NO:04

PROBLEM 4 : FIND THE TIME COMPLEXITY USING COUNTER METHOD

PROBLEM STATEMENT:

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

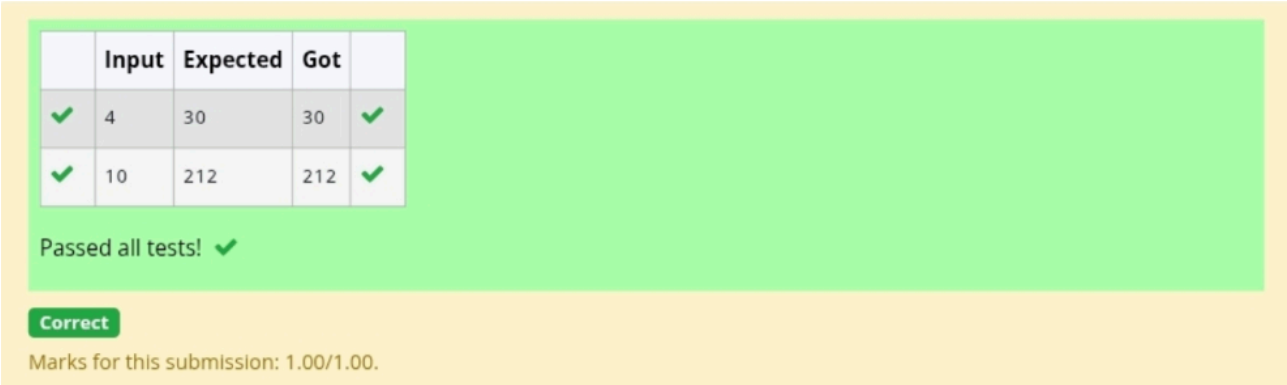
Print the value of the counter variable

CODE:

```c
#include<stdio.h>
int ct = 0;
void function(int n){
    int c=0;
    ct++;
    for(int i=n/2;i<n;i++){
        ct++;
        for(int j=1;j<n;j=2*j){
            ct++;
            for(int k=1;k<n;k=k*2){
                ct++;
                c++;
                ct++;
            }
            ct++;
        }
        ct++;
    }
    ct++;
}
int main(){
    int n;
    scanf("%d", &n);
    function(n);
    printf("%d", ct);
}
```

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OUTPUT:



	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

RESULT:

The code is executed successfully.